

Matin Ebrahimkhani

Curriculum Vitae



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RESEARCH INTERESTS

- Natural Language Processing:** Large Language Models, Language Understanding and Generation, Text Correction, Sequence Labeling, Multilingual NLP.
- Speech and Multimodal Language Systems:** Automatic Speech Recognition, Text-to-Speech Systems, Neural Voice Generation, Language-Speech Integration, and Multimodal Learning.
- Efficient and Trustworthy AI:** Model Adaptation and Fine-Tuning, Efficient Training and Inference, Evaluation of AI Systems, and Robust Machine Learning.
- Applied Machine Learning Systems:** Scalable AI Pipelines, Real-World Deployment of Machine Learning Models, and Reproducible AI Research.

EDUCATION

- 2020-24 **M.Sc. in Computer Engineering - Artificial Intelligence and Robotics**
Department of Computer Science and Information Technology
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
- GPA: 3.11/4 (15.54/20).**
 - Thesis: Correction of Space and ZWNJ in Persian Text, using Sequence Labeling with Pre-trained Large Language Models**
 - Mentor: Dr. Ebrahim Ansari
 - Main Courses: DL, ANN, ML, Data Mining, Statistical Pattern Recognition
- 2016-20 **B.Sc. in Information Technology Engineering**
Department of Computer Science and Information Technology
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran
- GPA: 3.25/4 (16.44/20).**
 - Thesis: Semi Automated Forex Assistant**
 - Mentor: Dr. Ebrahim Ansari
 - Main Courses: ML, NLP, Statistical Models, Intelligent Systems Design

PUBLICATIONS AND CONFERENCES

PerSpaCor: Correcting Space and ZWNJ Errors in Persian Text with Transformer Models

Matin Ebrahimkhani, Ebrahim Ansari

Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing (RANLP 2025)

- Novel Research:** Engineered and evaluated four original transformer-based classifier architectures—ReLU-Norm, DualStep DropNet, and SimplexNet—achieving a **State-of-the-Art macro F1-score of 97.26%** (98.38% with interactive correction).
- Dataset Leadership:** Independently curated, preprocessed, and **manually annotated** a massive dataset of over **12.7 million words**, creating a new high-quality benchmark for Persian NLP.
- Open Source Impact:** Released all SOTA models, code, and the annotated corpus to the research community and deployed a public web-based correction system to make the research accessible to end-users.
- Paper:** ACL Anthology – 2025.ranlp-1.40
- Artifacts:** Hugging Face/PerSpaCor | GitHub/PerSpaCor_components

PLANNED PUBLICATIONS

Efficient Adaptation of Foundation Models: A Comparative Study of Parameter-Efficient Fine-Tuning Methods for Domain-Specific Tasks

Matin Ebrahimkhani (first author), ...

In preparation

Investigating parameter-efficient fine-tuning techniques (LoRA, QLoRA, Adapters, Prefix Tuning, and full fine-tuning) across multiple NLP tasks (text classification, question answering, sequence labeling, text generation). The study will evaluate accuracy/F1 (for classification/QA/labeling), BLEU/ROUGE (for generation), and resource metrics (training time/GPU memory, inference latency, and number of trained parameters) to compare performance-efficiency trade-offs. Planned code and scripts will be released for reproducibility.

HONORS AND ACHIEVEMENTS

- 2024 **Ranked 58th Nationwide**
National Ph.D. Entrance Examination (Concours)
- 2022 **LLM Research Grant Recipient**
Awarded funding to support Large Language Model research
- 2021 **Founder of IASBS Music Society**
Established the student music community at the IASBS
- 2020 **Ranked 21st Nationwide**
National Graduate Entrance Examination (Concours)
M.Sc. admission in Computer Sciences
- 2019 **1st Place — IASBS ACM Programming Competition**
Member of the first competitive programming team at IASBS

SKILLS

Programming Languages

PYTHON	Expert — ML, NLP, and automation
C/C++	Advanced — systems and performance
BASH	Proficient — scripting and automation
JAVASCRIPT	Familiar — web development
HTML/CSS	Advanced — web design and development
SQL	Familiar — database operations
C#	Familiar — application development

Machine Learning & Deep Learning

PYTORCH	Advanced — research and production
TENSORFLOW	Intermediate — prototyping
HUGGINGFACE	Advanced — transformers, fine-tuning datasets and nlp pipelines
CUDA	Intermediate — GPU computing
TENSORBOARD	Model monitoring and analysis

Software Engineering & Tools

LINUX	Daily driver and server management
GIT	Proficient — VC and collaboration
DOCKER	Proficient — containerization
KUBERNETES	Intermediate — service management
TEX	Expert — academic writing

Mathematical Foundations

MACHINE LEARNING	Modeling and evaluation
DEEP LEARNING	Neural architectures and optimization
ALGORITHMS	Design and analysis
OPTIMIZATION	Applied mathematical methods
PROBABILITY THEORY	Statistical modeling
LINEAR ALGEBRA	Mathematical foundations

Efficient Adaptation of Foundation Models: A Comparative Study of Parameter-Efficient Fine-Tuning Methods for Domain-Specific Tasks

Matin Ebrahimkhani Mina Sirinchi, Mo Nasher, Mobin Ebrahimkhani
(In Progress)

Investigating parameter-efficient fine-tuning techniques (LoRA, QLoRA, Adapters, Prefix Tuning, and full fine-tuning) across multiple NLP tasks (text classification, question answering, sequence labeling, text generation). The study will evaluate accuracy/F1 (for classification/QA/labeling), BLEU/ROUGE (for generation), and resource metrics (training time/GPU memory, inference latency, and number of trained parameters) to compare performance-efficiency trade-offs. Planned code and scripts will be released for reproducibility.

RESEARCH AND WORK EXPERIENCE

REEICO

SEP 2024 – APR 2025

AI Researcher and Developer

Designed and developed an end-to-end **AI-powered dubbing system** for translating and localizing English video content into Persian. The pipeline integrated automatic speech recognition (ASR), **neural machine translation** (NMT), **text-to-speech synthesis** (TTS), and speaker-aware voice cloning. Implemented speaker diarization and alignment techniques. Independently handled the **MLOps** infrastructure, including model training pipelines, version control, containerization, and scalable deployment for reproducible and maintainable research.

JUL 2021 – SEP 2024

Freelance

Freelance Developer and Researcher

Designed and implemented web-based services and AI-driven applications across various domains. Conducted independent research on reinforcement learning and algorithmic trading, developing custom trading algorithms and indicators for the Forex market. Emphasized real-time data processing, predictive modeling, and the integration of machine learning into financial decision-making systems.

APR 2020 – JUL 2021

SAFA Corp.

Software Engineer and Data Analyst

Developed semi-automated and fully-automated robots for the Forex market. Statistical analysis on financial markets. Designend and implemented Machine learning algorithms on financial market data.

DEC 2018 – OCT 2019

Naghil Noghouli
Programmer

Programmer and tech team lead in the development of Naghil Noghouli game-app. Naghil Noghouli is a storytelling game-app for kids aged 3 – 12. Developed with Unity game engine and C# programming language.

JUL 2018 – SEP 2019

Chakavak Music Academy
Programmer

Implemented a learning management system (LMS) for Chakavak Music Academy, Zanjan, Iran.

OTHER INTERESTS

Free and Open-Source Software (FOSS), Computer Hardware and PC Building, DIY Home Projects, Canines, Percussion Instruments, Overlanding & Camping, Traveling, Cinema, Discovering Exotic Music, Poetry, and Fixing Things That Weren't Technically Broken.

Languages

ENGLISH	Near-native
PERSIAN	Native
AZERBAIJANI	Native
TURKISH	Elementary

TEACHING ASSISTANT EXPERIENCE

WINTER 2022 **Deep Learning**

Institute for Advanced Studies in Basic Sciences

Teaching Assistant; explaining deep learning concepts, designing and grading assignments.

FALL 2021 **Practical Introduction to Computer Science**

Institute for Advanced Studies in Basic Sciences

Teaching Assistant; conducting problem-solving sessions, designing and grading programming assignments.

SPRING 2020 **Information Retrieval**

Institute for Advanced Studies in Basic Sciences

Teaching Assistant; grading programming and written assignments.

WINTER 2020 **Advanced Programming II**

Institute for Advanced Studies in Basic Sciences

Teaching Assistant; conducting exercise sessions, explaining concepts, and designing and grading assignments.

FALL 2019 **Advanced Programming I**

Institute for Advanced Studies in Basic Sciences

Teaching Assistant; conducting exercise sessions, explaining concepts, and designing and grading assignments.

FALL 2018 **Digital Logic Circuits**

Institute for Advanced Studies in Basic Sciences

Teaching Assistant; grading programming and written assignments.

REFERENCES

Dr. Ebrahim Ansari

Assistant Professor, Dept. of Computer Science & IT, IASBS, Zanjan, Iran
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Dr. Zahra Narimani

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Dr. Parvin Razzaghi

Assistant Professor, Dept. of Computer Science & IT, IASBS, Zanjan, Iran
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